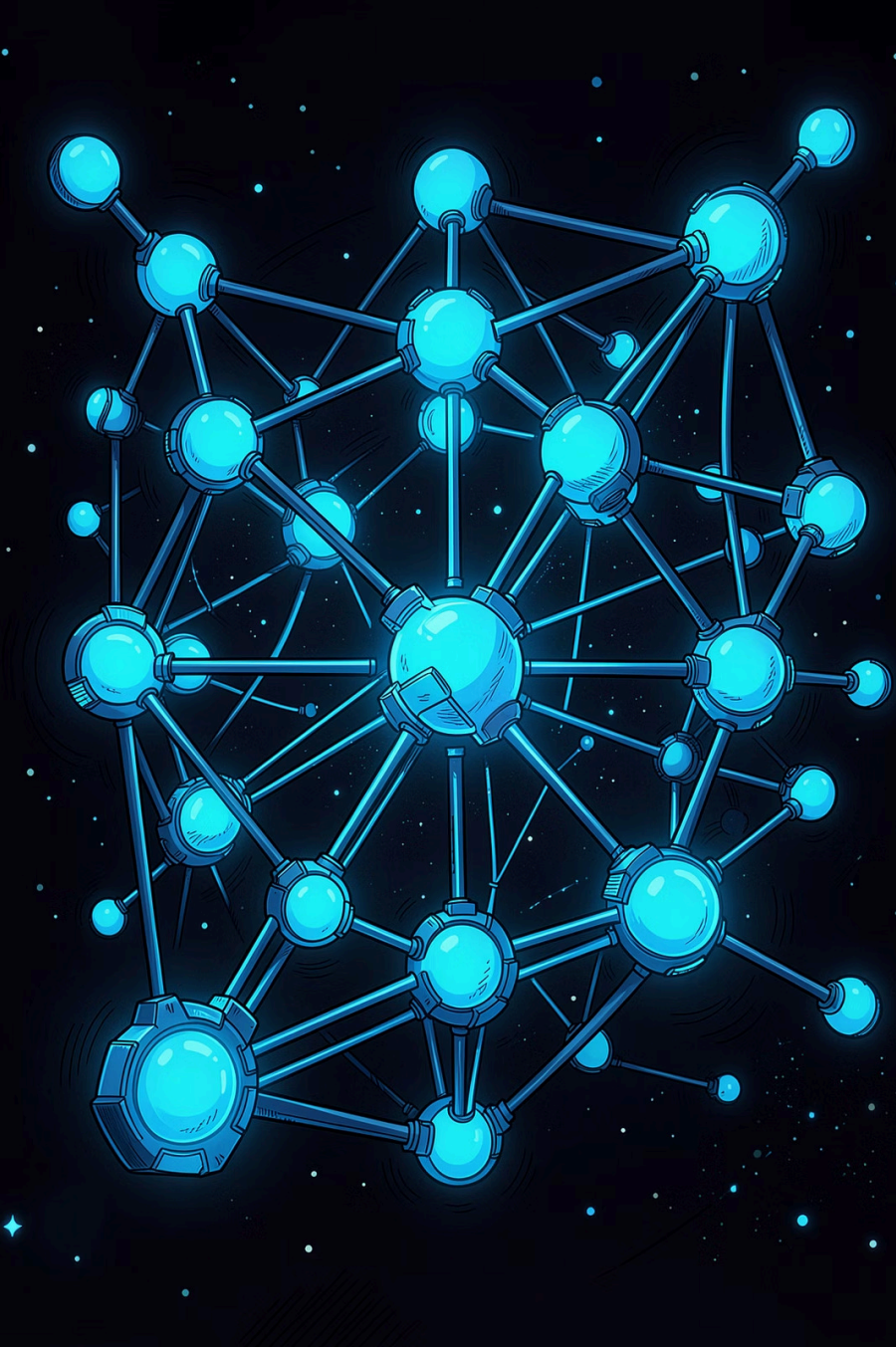




Building a Gamified 3D Interface for Neosentience Research

UI Design & Research Interface Development

Duke University Emergence Lab

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What Is This Project?

Researchers synthesized **over 1,000 papers** across AI, neuroscience, robotics, computation, philosophy, and consciousness. Traditional search interfaces make it nearly impossible to explore relationships across these disciplines. This project creates a **3D gamified world** where users naturally discover knowledge through spatial exploration — and my role was to design and build the interface that makes it possible.

1

1,000+ Papers

Synthesized research corpus

2

Knowledge Graph

Structured relationships

3

3D Gamified World

Spatial exploration

4

User Discovery

Natural knowledge navigation

Knowledge Organization

The research corpus is organized into **over 80 conceptual categories**, grouped into major interdisciplinary domains. This taxonomy forms the structural backbone of the 3D world — each zone maps to a research domain, giving users a spatial sense of where ideas live and how they connect.



Interface Design Goals

Rather than starting with features, the design was driven by five core objectives that shaped every interface decision.

1

Accessible Exploration

Make 1,000+ papers navigable without overwhelming users

2

Non-Linear Discovery

Encourage serendipitous browsing rather than linear keyword search

3

Visualize Relationships

Show how concepts connect across disciplines

4

Support Experiments

Enable behavioral research data collection through interface interactions

5

Engaging Experience

Provide a game-like environment that sustains motivation

The User Journey

Before diving into individual features, here's the complete workflow — from first login to ongoing discovery. This mental model grounds every design decision that follows.



Each phase is instrumented to capture behavioral data, turning every user session into a research opportunity.

Navigating the 3D World



The interactive world map is the primary navigation layer. Users zoom, pan, and click through labeled research zones — transforming abstract literature into a spatial landscape they can physically explore.

Interactive World Map

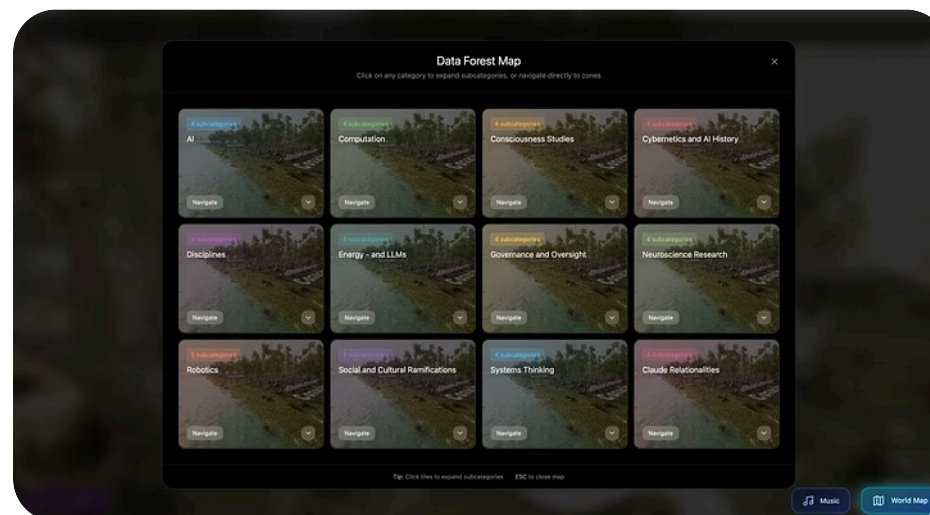
Zoom, pan, and click through research zones

Zone Labels & Controls

Clear region markers and map navigation tools

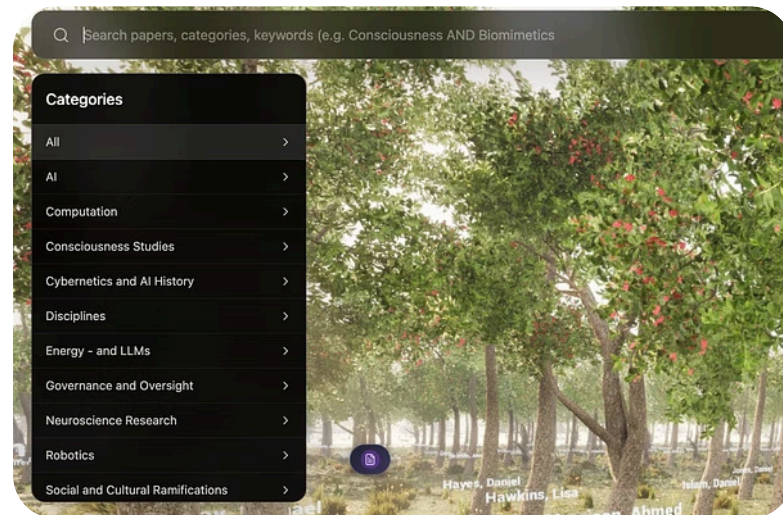
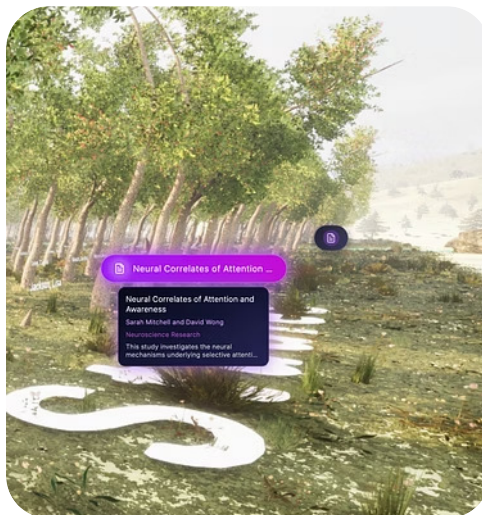
Behavioral Tracking

Navigation patterns, visited regions, and interaction frequency are logged for research



Finding Research

Users retrieve papers through **multiple discovery methods** — not just traditional keyword search. The interface supports several search modes and an association selector, enabling researchers to compare search strategies, measure efficiency, and evaluate discovery behavior across participants.



Search Bar & Filters

Keyword search with configurable filters



Search Modes

Multiple discovery pathways beyond linear search



Association Selector

Choose how concepts are linked during retrieval

Building Relationships Between Ideas

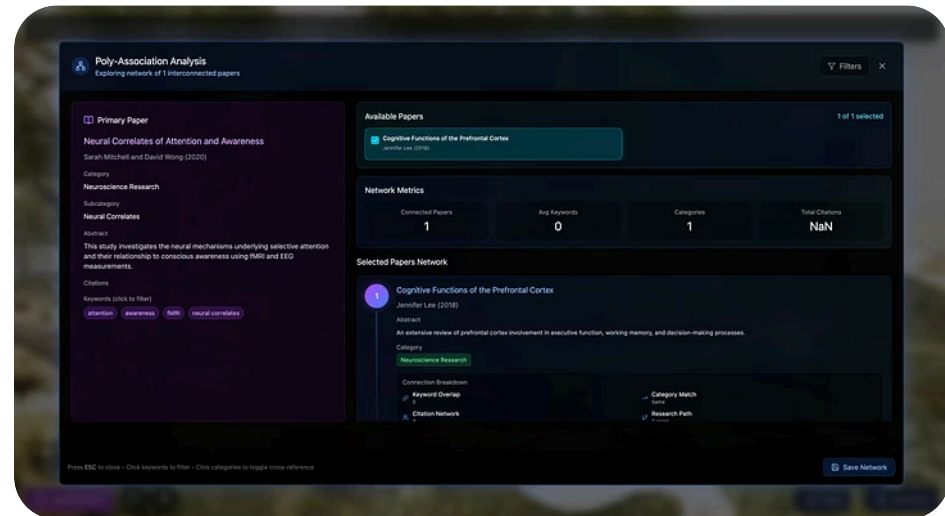
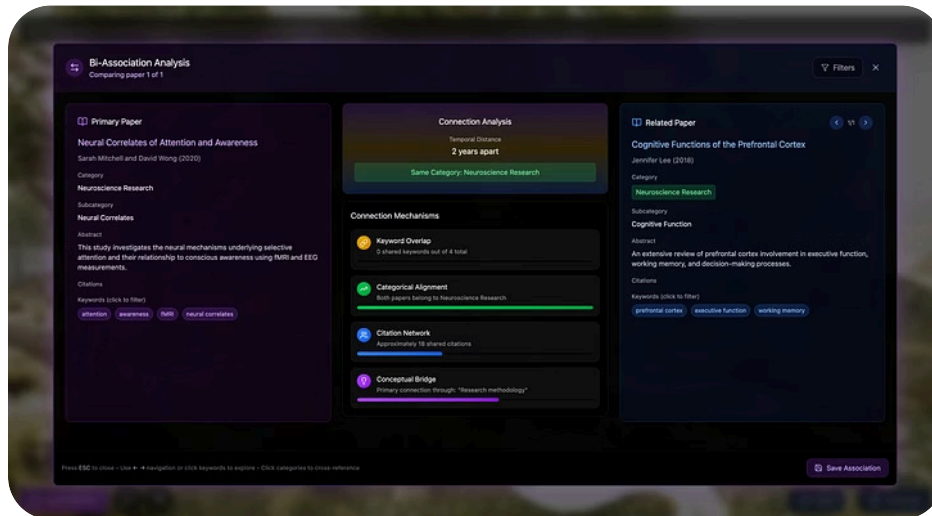
This is the interface's most distinctive research contribution. Instead of passively reading papers, users **actively construct relationships** between concepts — enabling the system to observe reasoning patterns, capture knowledge construction, and analyze association strategies.

Bi-Association View

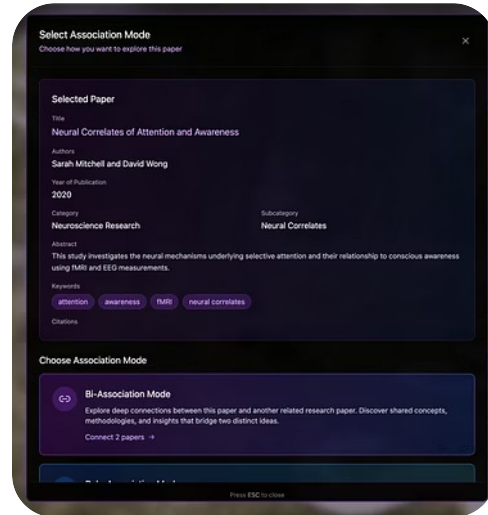
Connect two concepts and explore their relationship

Poly-Association View

Build multi-node networks of interconnected ideas



Exploring Individual Papers



Each paper is represented as an interactive node. Users can inspect full metadata, revisit past discoveries, and organize relevant literature into personal collections — turning the interface into both a research tool and a data collection instrument.

→ **Interactive Paper Nodes**
Click any node to open the detail panel

→ **Metadata Inspection**
View authors, abstracts, and categorization

→ **Bookmarks & Saved Papers**
Build a personal library of relevant literature

→ **Revisit Discoveries**
Return to previous findings at any time

Research-Oriented Interface Features

Adaptive Chatbot

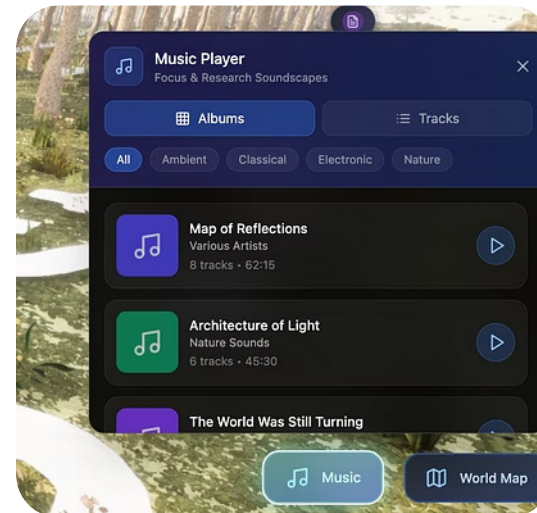
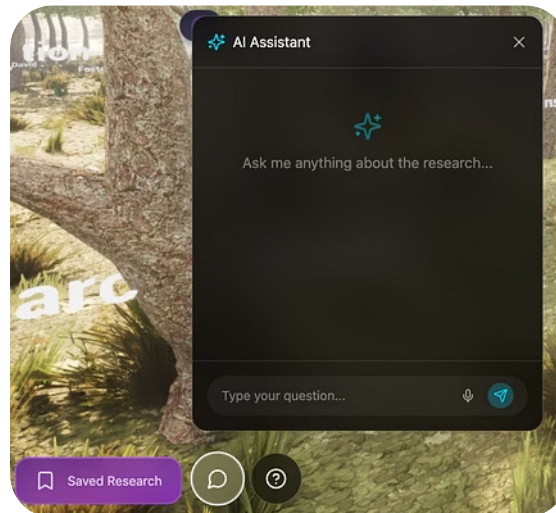
Contextual assistance that responds to help requests and supports conversational exploration of the research space.

- Adaptive assistance
- Help request handling
- Conversational discovery

Experimental Music Player

An integrated music player provides users with optional background music while exploring the 3D research environment.

- Provides a more immersive experience
- Allows users to listen to music while exploring
- Improves comfort during extended research sessions
- Offers convenient playback controls without leaving the application



Help System

On-demand guidance for participants

Participant Authentication

Secure login and session management

Reusable UI Components

Modular design system for consistency

Implementation

Building a robust and scalable interface for complex research demands meticulous technical execution. Our approach focused on modularity, performance, and detailed behavioral instrumentation.

Key Interfaces Developed

- Login & Authentication
- Dynamic 3D World Map
- Advanced Search & Filters
- Bi- & Poly-Association Views
- Detailed Paper Panels
- Adaptive Chatbot Integration
- Saved Papers & Collections
- Reusable UI Component Library

Technical Backbone

The entire system is implemented using **TypeScript** and **React**, leveraging a comprehensive library of reusable UI components for consistency and rapid development.

Critical research functionality involved:

- Precise interaction logging
- Detailed click event tracking
- Drag-and-drop behavioral traces
- Comprehensive user session analytics

My Contributions

My involvement spanned the entire project lifecycle, from foundational research to interface delivery, focusing on creating a truly unique and impactful research tool.



UI Development

Designed and implemented over **10 bespoke interfaces** and **50+ interactive features**, establishing the core functionality of the immersive 3D research environment.



Research Synthesis

Synthesized insights from over **1,000 complex research papers**, structuring content into **80+ interdisciplinary categories** to form the system's knowledge backbone.



User Experience

Crafted an intuitive and engaging exploratory interface, translating advanced AI research into accessible, game-like interactions for behavioral study.

Impact & Future Work

Our journey doesn't end here. The initial prototype lays the groundwork for continuous evolution, with exciting opportunities to expand its capabilities and deepen its research insights.



Expand Paper Database

Continuously grow the repository of interdisciplinary research for broader coverage.



Improve Visualization

Enhance the 3D environment for clearer concept mapping and intuitive navigation.



AI Recommendations

Integrate intelligent systems to suggest relevant papers and novel connections.



Evaluate User Studies

Conduct formal studies to measure the interface's impact on discovery and learning.



Collaborative Exploration

Develop features for multiple users to explore and build knowledge together.